

Edinburgh Lectures in Language Evolution 2023

Wednesday 30th, also 6th, 13th, 20th November

4pm, 7 George Square, Lecture Theatre F21

This week's speaker: Shira Tal, Edinburgh

Investigating the developmental trajectory of cognitive biases that shape language

One of the central questions in the language sciences is how the world's languages are impacted by our cognition. A host of factors have been proposed to shape language patterns, among them constraints on how we learn and communicate. However, very few empirical studies have investigated how these cognitive biases might differ in adults and children, who are fundamentally different types of language learners and users. This gap in research is crucial, as understanding these differences can inform longstanding debates about who drives language change. Moreover, studying young children provides a unique window to our early cognition, allowing for a better understanding of the nature of the cognitive biases themselves. In this talk, I will present several studies investigating in young children key cognitive biases that are hypothesized to shape language over time.

Origins and Evolution of Language

Week 7: The cultural evolution of language

Kenny Smith

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Plan for today

- Rattle through a few questions on last week's content
- Cultural evolution of language
 - Questions from the reading quiz
 - Uniformitarianism
 - Learning, use, and language change
 - Cultural transmission and the evolution of symbols
 - Cultural transmission and the evolution of structure

A question from the pre-lecture quiz

“To what extent do you think Theory of Mind evolved according to biological or cultural evolution?”

A question from the pre-lecture quiz

“Are cooperation and cultural transmission, and cultural evolution and cultural transmission two different arguments in the evolution of language? Do these theories differ based on the focus on pragmatic competence and required mind reading abilities in social groups on the one hand, and cultural evolution and selective pressures on the linguistic system to become more learnable and generalisable on the other?”

An unwelcome question on subdoxastic states

“At what point does a subdoxastic state stop being some sort of genetic inexplicable instinct and start becoming something that is not quite a thought or belief but can interact with thoughts and influence behavior? It seems as though the line is quite blurred between subdoxastic states and innate knowledge and actions if we cannot integrate a subdoxastic state with our other knowledge and it is not a full belief”

Food for thought from the pre-lecture quiz

“Inspired by this article, it reminds me of integration between humans and artificial intelligence, especially in Chatgpt times. Do you think machines need to be able to speculate human mental states in order to interact naturally with humans? Or is it possible for machines to develop an efficient way of communicating that does not rely on theory of mind?”

“In an advanced technological society (especially with the rise of AI mediated infrastructures), could one suspect that we may reach a point where it may become harder or easier to detect the false beliefs of others?”

What you've seen so far (1/2)

Human linguistic communication has unusual properties (see week 3)

- Evidence for 2nd order intentionality in communication is rare in other animals
- Lots of structured communication out there, but structure is simpler and typically not meaning-related

Human capacity to sustain complex non-linguistic cultures (e.g. tools) is also unusual (see week 4)

- Language implicated in maintenance of stone tool technology?

What you've seen so far (2/2)

Human capacity for learning complex meaning-bearing communicative signals is unusual (see weeks 5-6)

- Vocal learning seen in other animals, but limited in our closest relatives?
- Other animals can learn sequencing constraints, but only simple ones have been tested
- Other animals can learn rules of meaningful combination, but few systematic studies

Human motivation to share mental states and aptitude to reason about the mental states of others is unusual (weeks 3 and 6)

- Mitteilungsbedürfnis is weird!
- Evidence for 2nd order intentionality in communication is rare in other animals
- Some evidence of capacity to reason about knowledge, ignorance and false belief in other apes, but only in competitive contexts
- Complex social living a possible selection pressure driving human brain expansion?

The human package

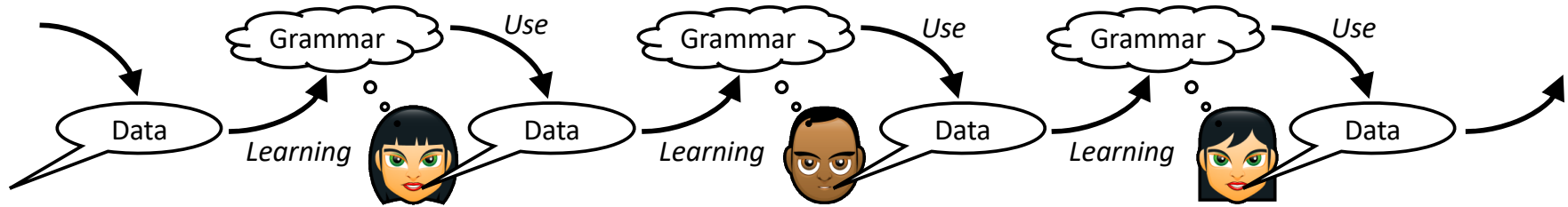
Somehow, we ended up with

- The ability to learn complex grammars
 - capacity for complex vocal imitation
 - ability to learn complex sequencing constraints
 - ability to learn compositional meaning-form mappings
- The ability and motivation to mindread and mindshare

This sets up the preconditions for the **cultural transmission of learned, meaning-bearing communication**

- Once that's in place, exciting stuff happens

The cultural evolution of language



- Language is passed from person to person by **learning**
- People learn from language as it is **used in communication**
- Language **evolves** in response to its learning and use

Uniformitarianism (in geology)

James Hutton (1726-1797)



Uniformitarianism: the present is the key to the past

“from what has actually been, we have data for concluding with regard to that which is to happen thereafter.”



John Playfair
(1748-1819)

“On us who saw these phenomenon for the first time the impression will not easily be forgotten...We felt necessarily carried back to a time when the schistus on which we stood was yet at the bottom of the sea, and when the sandstone before us was only beginning to be deposited, in the shape of sand or mud, from the waters of the supercontinent ocean... The mind seemed to grow giddy by looking so far back into the abyss of time; and whilst we listened with earnestness and admiration to the philosopher who was now unfolding to us the order and series of these wonderful events, we became sensible how much further reason may sometimes go than imagination may venture to follow.””



Lyell on catastrophism



Charles Lyell (1797-1875)

“Never was there a doctrine more calculated to foster indolence, and to blunt the keen edge of curiosity, than this assumption of the discordance between the former and the existing causes of change... The student was taught to despond from the first. Geology, it was affirmed, could never arise to the rank of an exact science... [With catastrophism] we see the ancient spirit of speculation revived, and a desire manifestly shown to cut, rather than patiently untie, the Gordian Knot”

Lyell, C. (1854). *Principles of Geology: Being an Attempt to Explain the Former Changes of the Earth's Surface, by Reference to Causes Now in Operation*

Uniformitarianism in evolutionary linguistics

The present is the key to the past

The more we can explain in terms of processes we can observe in the present day, the happier we should be

- Learning and use explain language change visible in the present and the recent historical record
- Can we explain (some of) language origins in terms of the same processes?
- Rather than catastrophism, e.g. language evolved in a single dramatic step due to some single magical event or macromutation

Importantly, uniformity of **process**, not of state: we don't have to say languages have always looked as they do now! (see e.g. Heine & Kuteva, 2002)

Language change

Language change (as attested in the historical record / inferable from synchronic data) is a consequence of:

- Speakers trying to convey meaning efficiently
- Hearers trying to infer speaker meaning
- Language learners (and everyone else) seeking regularities in the linguistic data they encounter

These processes are inherent to the transmission of language via learning and (ostensive-inferential) use

Ad-hoc extension to meet communicative needs



tick-tock!



axe



corkscrew?



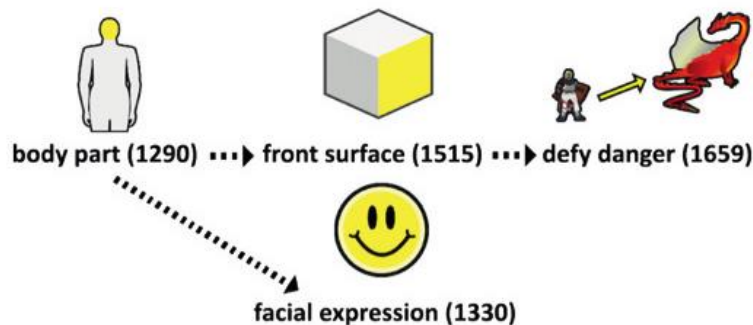
The **beard** is still waiting for his spaghetti



You are **my fire**
The one desire
Believe when I say
I want it that way



“A reef of dead metaphors” (Deutscher, 2005)



From Ramiro, C., Srinivasan, M., Malt, B. C., & Yu, X. (2018). Algorithms in the historical emergence of word senses. *Proceedings of the National Academy of Sciences, USA*, 115, 2323-2328.

“She was *thrilled* to *discover* that the *assessment board* had *decided* to make her *rival redundant*”

thrill: from thirl, “to pierce”

discover: remove the cover from

assessment: from *assidere*, “to sit by” (in judgment)

board: plank

decided: from *de-caedere*, “cut off”

rival: from *rivalis*, someone who shares the same river

redundant: from *redundantem*, “overflow”

From p. 125 of Deutscher, G. (2005). *The Unfolding of Language*. New York, NY: Picador.

Grammaticalization

E.g.: development of future tense markers from verbs of motion

I am going to Gothenburg

MOTION

I am going to stay at home

INTENTION

It is going to rain

FUTURE

Grammaticalization

E.g.: development of future tense markers from verbs of motion

I am going to Gothenburg

MOTION (+ INTENTION)

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INTENTION

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FUTURE

Grammaticalization

E.g.: development of future tense markers from verbs of motion

I am going to Gothenburg

MOTION (+ INTENTION)

I am going to buy you a gift!

MOTION + INTENTION

I am going to stay at home

INTENTION

It is going to rain

FUTURE

Grammaticalization

E.g.: development of future tense markers from verbs of motion

I am going to Gothenburg

MOTION (+ INTENTION)

I am going to buy you a gift!

MOTION + INTENTION

I am going to stay at home

INTENTION (+ FUTURE)

It is going to rain

FUTURE

Grammaticalization

E.g.: development of future tense markers from verbs of motion

<i>I am going to Gothenburg</i>	MOTION (+ INTENTION)
<i>I am going to buy you a gift!</i>	MOTION + INTENTION
<i>I am going to stay at home</i>	INTENTION (+ FUTURE)
<i>I am going to stay at home tomorrow</i>	INTENTION + FUTURE
<i>It is going to rain</i>	FUTURE

Grammaticalization

E.g.: development of future tense markers from verbs of motion

I am going to Gothenburg

MOTION (+ INTENTION)

I am going to buy you a gift!

MOTION + INTENTION

I am going to stay at home

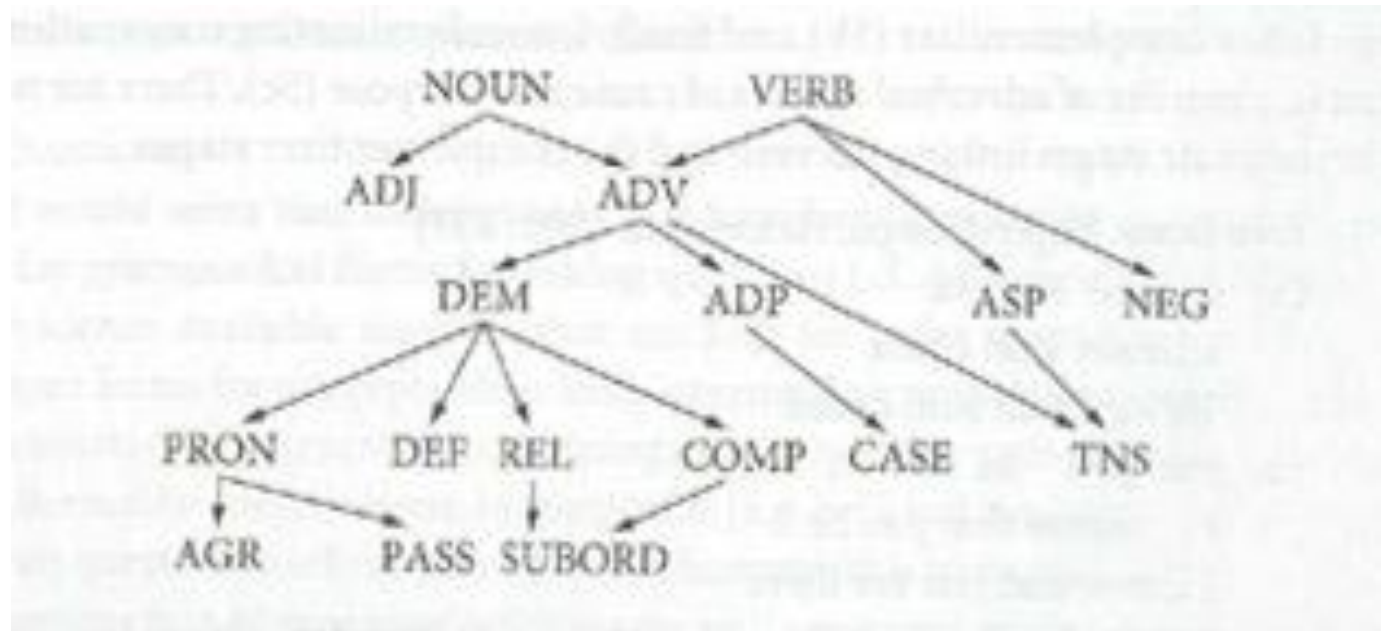
INTENTION (+ FUTURE)

I am going to stay at home tomorrow

INTENTION + FUTURE

It's gonna to rain

FUTURE



nature

WORDS ON THE BRINK

The evolution
of language

NUCLEAR WEAPONS

How to be an
IAEA inspector

ATMOSPHERIC HUMIDITY

The human touch

GENE SILENCING

Non-toxic RNA
inhibition

NATUREJOBS

Mentoring
awards

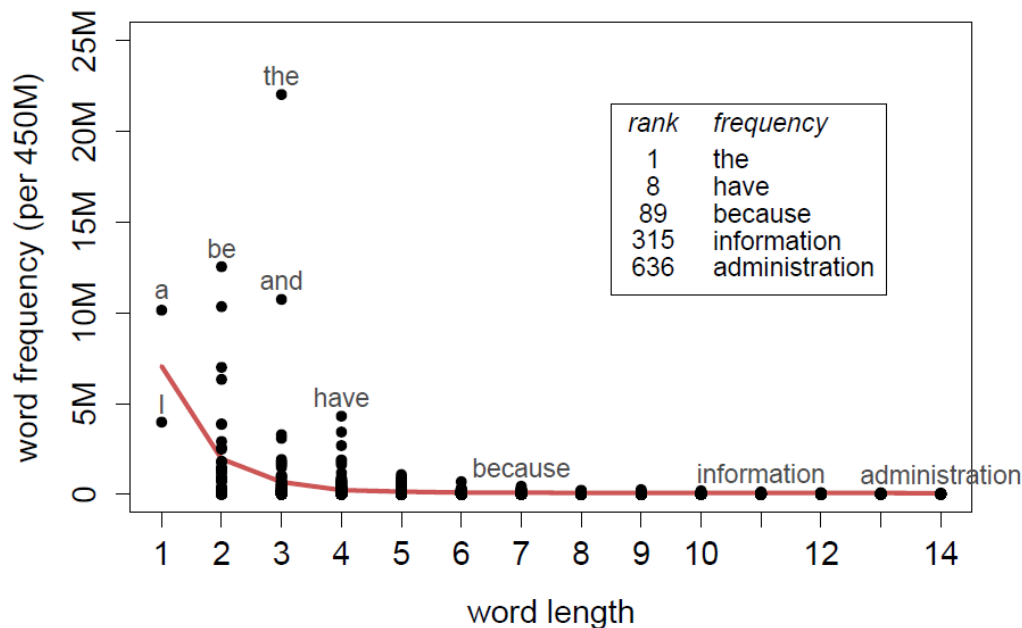


Analological extension & “system pressure”

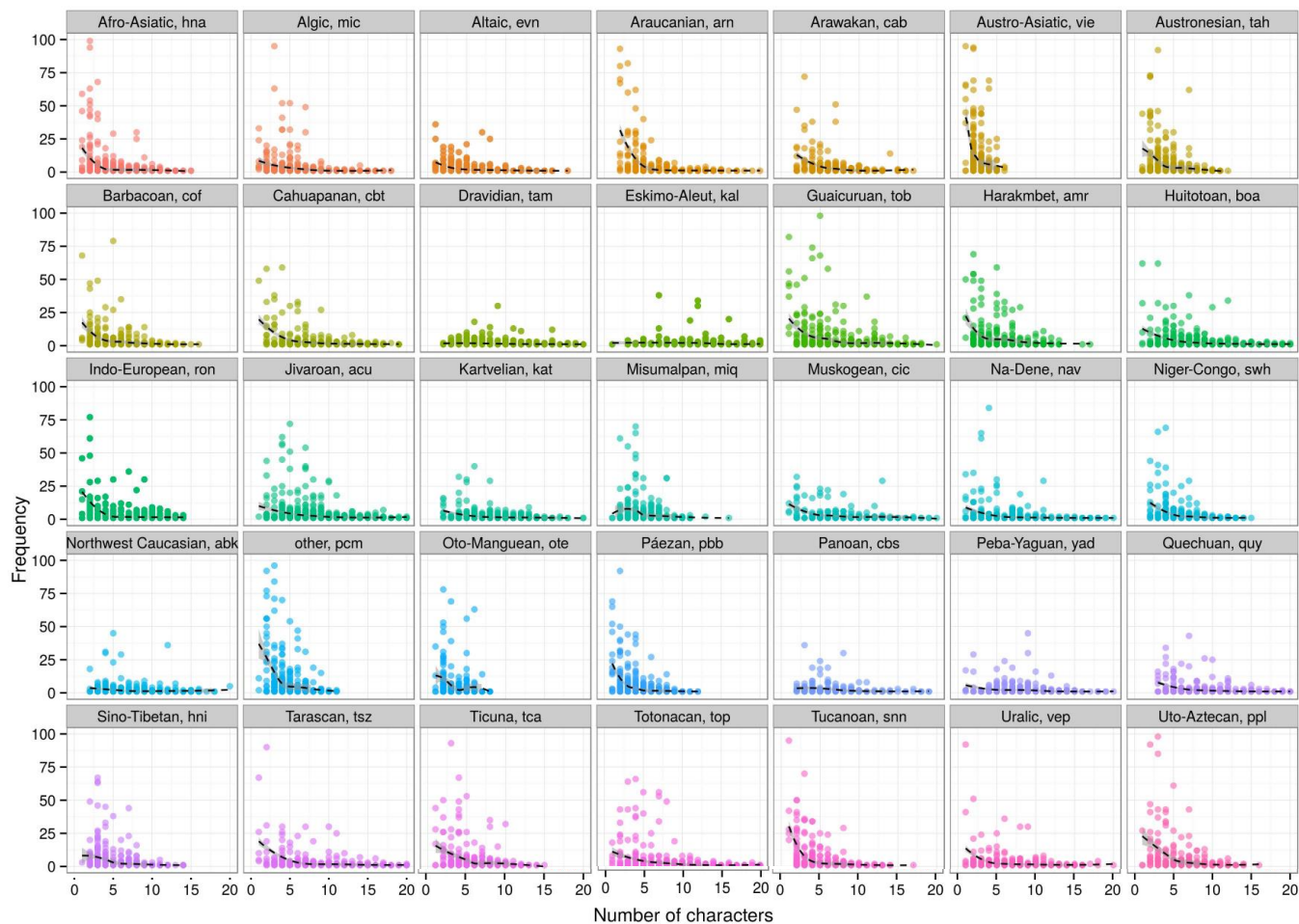
Cover of issue featuring Lieberman, E., Michel, J. B., Jackson, J., Tang, T., & Nowak, M. A. (2007). Quantifying the evolutionary dynamics of language. *Nature*, 449, 713-716.

Analogical extension & “system pressure”

Frequent words tend to be short (Zipf’s Law of Abbreviation)



From Kanwal, J., Smith, K., Culbertson, J., & Kirby, S. (2017). Zipf’s Law of Abbreviation and the Principle of Least Effort: Language users optimise a miniature lexicon for efficient communication. *Cognition*, 165, 45-52.



From Bentz, C., & Ferrer-i-Cancho, R. (2016). Zipf's law of abbreviation as a language universal. In Bentz, C., Jäger, G., & Yanovich, I. (Eds.) *Proceedings of the Leiden Workshop on capturing phylogenetic algorithms for linguistics*.

Analogical extension & “system pressure”

Frequent words tend to be short (Zipf’s Law of Abbreviation)

But system-level pressures favor **regularity**

TABLE 12.9. An unattested system

English	SG	PL	Percentage of singular	Hypothetical language
<i>house</i>	49295	9840	83	<i>house-Ø/house-ssss</i>
<i>hare</i>	488	136	78	<i>hare-Ø/hare-sss</i>
<i>bear</i>	1182	611	66	<i>bear-Ø/bear-ss</i>
<i>window</i>	9936	8506	54	<i>window-Ø/window-s</i>
<i>feather</i>	487	810	38	<i>feather-one/feather-Ø</i>
<i>parent</i>	3706	15956	19	<i>parent-oneone/parent-Ø</i>

From Haspelmath, M. (2014). On system pressure competing with economic motivation. In MacWhinney, B., Malchukov, A., & Moravcsik, E. (Eds) *Competing Motivations in Grammar and Usage* (pp. 197-208). Oxford: Oxford University Press.

Language change

Language change (as attested in the historical record / inferable from synchronic data) is a consequence of:

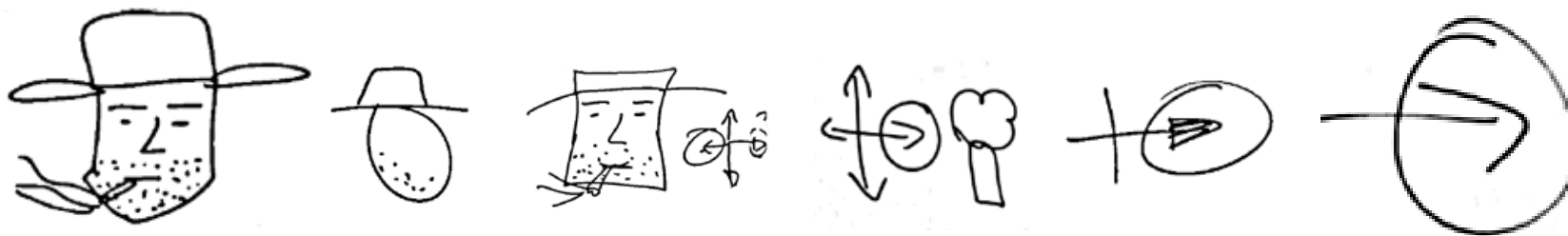
- Speakers trying to convey meaning efficiently
- Hearers trying to infer speaker meaning
- Language learners (and everyone else) seeking regularities in the linguistic data they encounter

These processes are inherent to the transmission of language via learning and (ostensive-inferential) use

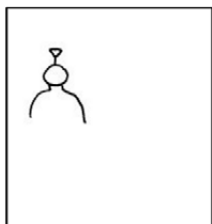
To what extent can these same processes explain the origins of fundamental properties of linguistic systems?

Example: the evolution of symbols

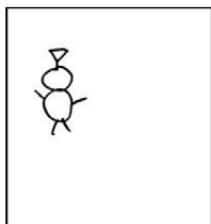
The evolution of arbitrary symbols in the lab (from week 1)



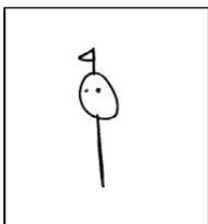
**Participant 1
(Round 1)**



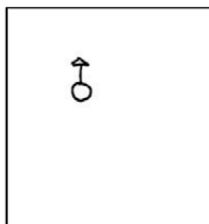
**Participant 2
(Round 2)**



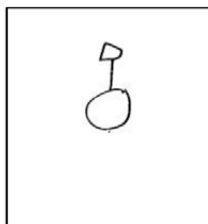
**Participant 3
(Round 3)**



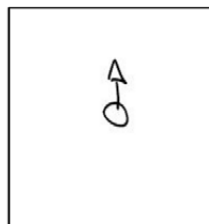
**Participant 4
(Round 4)**



**Participant 5
(Round 5)**



**Participant 6
(Round 6)**



Garrod, S. et al. (2007). Foundations of Representation: Where Might Graphical Symbol Systems Come From? *Cognitive Science*, 31, 961-987

Caldwell, C. A., & Smith, K. (2012). Cultural evolution and the perpetuation of arbitrary communicative conventions in experimental microsocieties. *PLoS ONE*, 7, e43807.

Example: the evolution of structure

Reminder: Language's communicative power comes from its **structure**

Compositionality: the meaning of an expression is a function of the meaning of its parts and the way in which they are combined

$S \rightarrow NP VP \quad VP'(NP')$

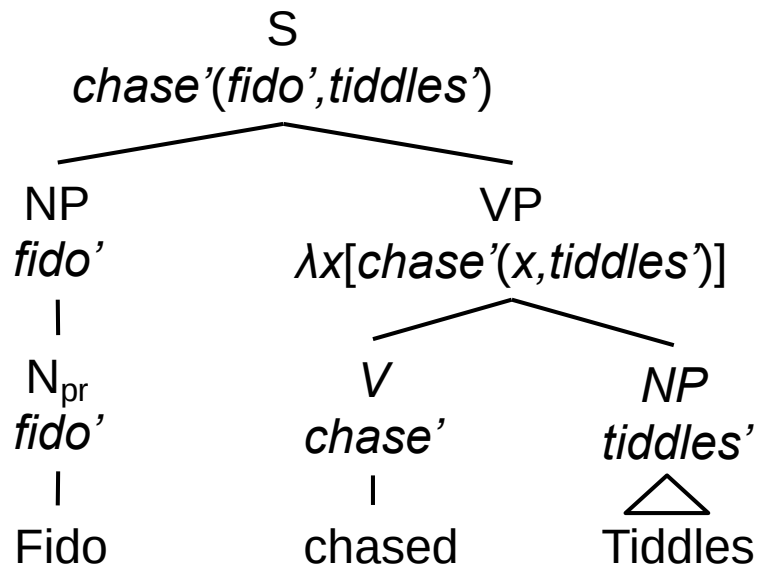
$NP \rightarrow N_{pr} \quad N'_{pr}$

$N_{pr} \rightarrow \text{Fido} \quad fido'$

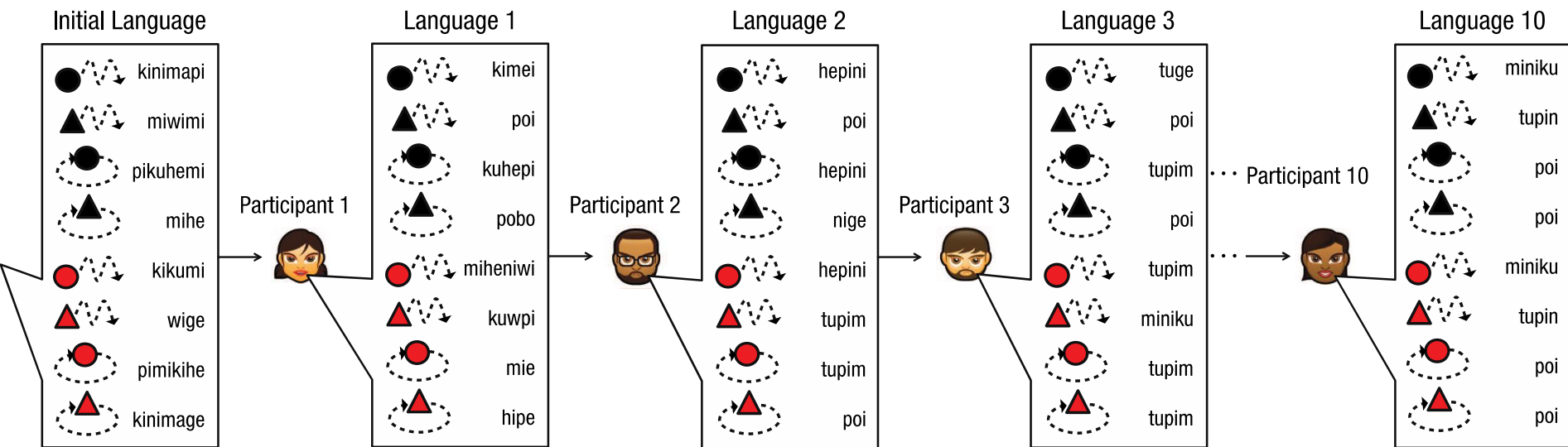
$N_{pr} \rightarrow \text{Tiddles} \quad tiddles'$

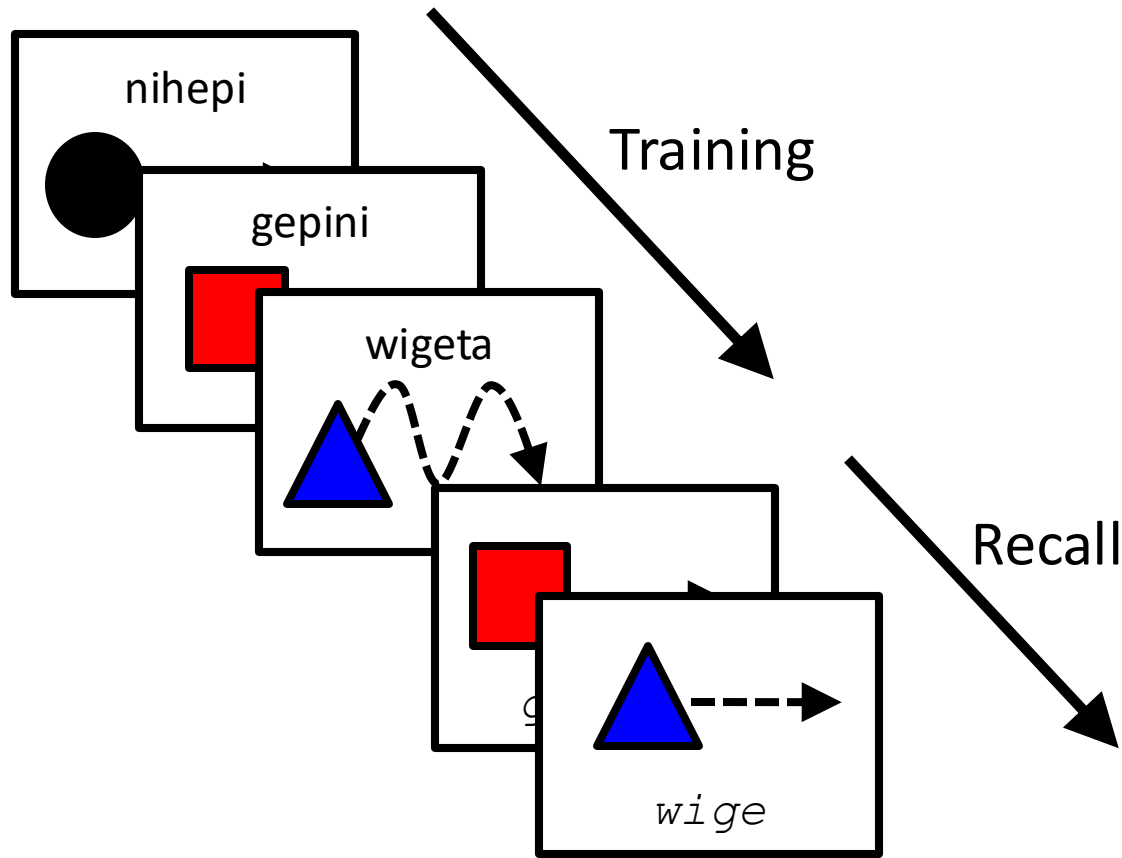
$VP \rightarrow V NP \quad V'(NP')$

$V \rightarrow \text{chased} \quad \lambda x [\lambda y [(chase'(x,y))]]$



Iterated Learning





Kirby, S., Cornish, H., & Smith, K. (2008). Cumulative cultural evolution in the laboratory: An experimental approach to the origins of structure in human language. *PNAS*, 105, 10681-10686.

An initial **holistic** language from chain 4

--→	wimaku	miniki	gepinini	□
	nihepi	wigemi	mahekuki	○
	wikima	nipikuge	hema	△
↻	miwiniku	pinipi	kihemiwi	□
	kinimapi	wikuki	kikumi	○
	miwimi	nipi	wige	△
↻	gepihemi	kunige	miki	□
	pikuhemi	kimaki	pimikihe	○
	mihe	winige	kinimage	△

Seen vs unseen

→			gepinini	□
			mahekuki	○
	wikima	nipikuge		△
↻	miwiniku		kihemiwi	□
		wikuki		○
	miwimi		wige	△
↻	gepihemi			□
	pikuhemi	kimaki		○
		winige	kinimage	△

Generation 1 language from chain 4

--→	nige	miniku	poh	□
	mip	mpo	miniku	○
	tuge	tuge	weg	△
↻	pemini	kupini	pon	□
	kimei	miwn	miheniw	○
	poi	mhip	kuwpi	△
↻	hepinimi	himini	hipe	□
	kuhepi	wige	mie	○
	pobo	tupim	hipe	△

Generation 2 language from chain 4

-->	nige	tuge	tuge	□
	nige	nige	tuge	○
	nige	tuge	mpo	△
↗↘	nige	tuge	mihenu	□
	hepini	miniku	hepini	○
	poi	mpo	tupim	△
↻	hepini	miniku	tupim	□
	hepini	tupim	tupim	○
	nige	tupim	poi	△

Generation 3 language from chain 4

--→	mihenu	tuge	tuge	□
	tuge	tuge	tuge	○
	nige	tuge	tuge	△
↗↘→	miniku	tuge	minihu	□
	tuge	tupim	tupim	○
	poi	tuge	miniku	△
↻	mihenu	tupim	tupim	□
	tupim	tupim	tupim	○
	poi	tupim	tupim	△

Generation 4 language from chain 4

--→	tuge	tuge	tuge	□
	tuge	tuge	tuge	○
		tuge	tuge	△
↗↘→	mihunu	tupim	miniku	□
	tupim	tupim	tupim	○
	poi	miniku	miniku	△
↻	tupim	tupim	tupim	□
	tupim	tupim	tupim	○
	poi	tupim	tupim	△

Generation 5 language from chain 4

-->	tuge	tuge	tuge	□
	tuge	tuge	tuge	○
	tuge	tuge	tuge	△
~>	minuhu	tupim	tupim	□
	tupim	tupim	tupim	○
	miniku	tupim	miniku	△
↻	tupim	tupim	tupim	□
	tupim	tupim	tupim	○
	poi	tupim	tupim	△

Generation 6 language from chain 4

-->	tuge	tuge	tuge	□
	tuge	tuge	tuge	○
	tuge	tuge	tuge	△
~>	miniku	tupin	tupim	□
	miniku	miniku	miniku	○
	miniku	tupin	tupin	△
↻	poi	tupin	tupim	□
	poi	poi	poi	○
	poi	tupin	tupim	△

Generation 7 language from chain 4

--→	tuge	tuge	tuge	□
	tuge	tuge	tuge	○
	tuge	tuge	tuge	△
↻	miniku	miniku	tupim	□
	miniku	miniku	miniku	○
	miniku	tupin	miniku	△
↻	poi	poi	tupim	□
	poi	poi	poi	○
	poi	tupin	poi	△

Generation 8 language from chain 4

--→	tuge	tuge	tuge	□
	tuge	tuge	tuge	○
	tuge	tuge	tuge	△
↻	tupim	tupim	tupim	□
	miniku	miniku	miniku	○
	tupin	tupin	tupin	△
↻	poi	poi	poi	□
	poi	poi	poi	○
	poi	poi	poi	△

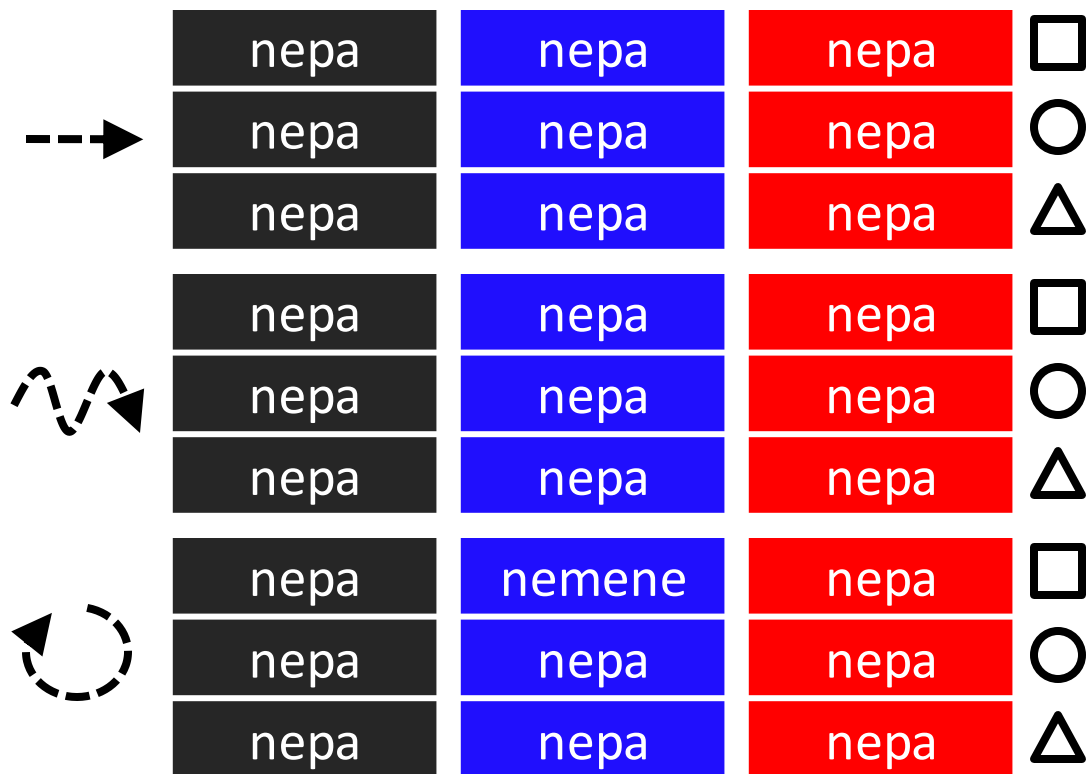
Generation 9 language from chain 4

-->	tuge	tuge	tuge	□
	tuge	tuge	tuge	○
	tuge	tuge	tuge	△
↗↘	tupim	tupim	tupim	□
	miniku	miniku	miniku	○
	tupin	tupin	tupin	△
↻	poi	poi	poi	□
	poi	poi	poi	○
	poi	poi	poi	△

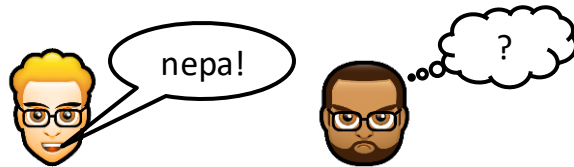
Generation 10 language from chain 4

--→	tuge	tuge	tuge	□
	tuge	tuge	tuge	○
	tuge	tuge	tuge	△
↻	tupim	tupim	tupim	□
	miniku	miniku	miniku	○
	tupin	tupin	tupin	△
↻	poi	poi	poi	□
	poi	poi	poi	○
	poi	poi	poi	△

Final language from chain 1 (!)



The languages become **degenerate**



Learnability and degeneracy

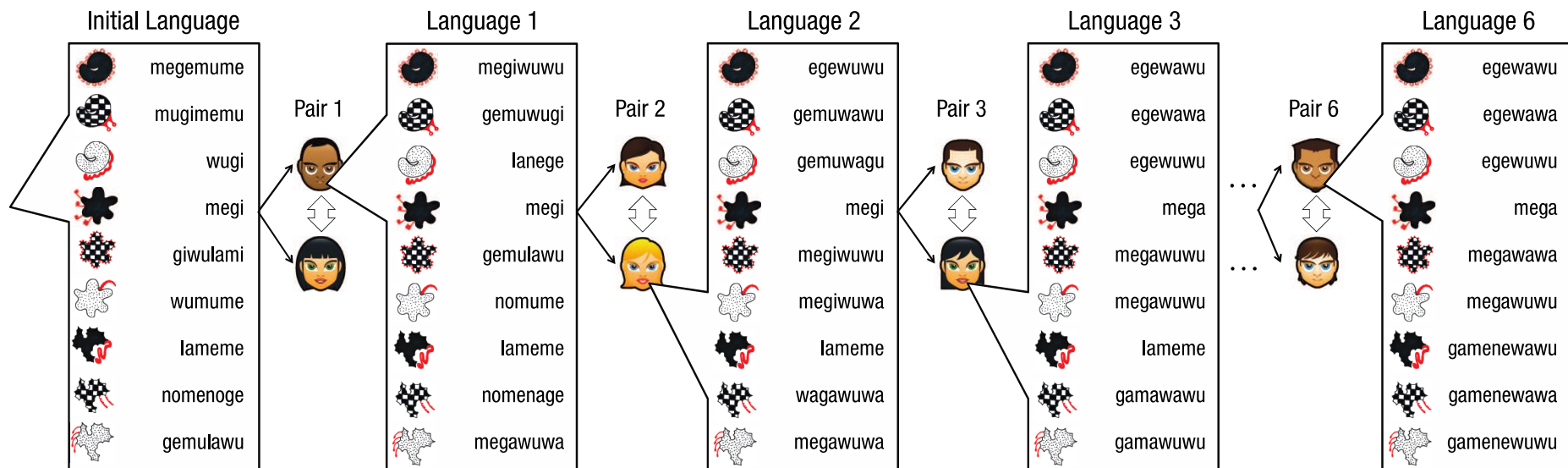
Learners prefer simpler languages

The only pressure in Kirby, Cornish & Smith (2008) Experiment 1 is **learnability**

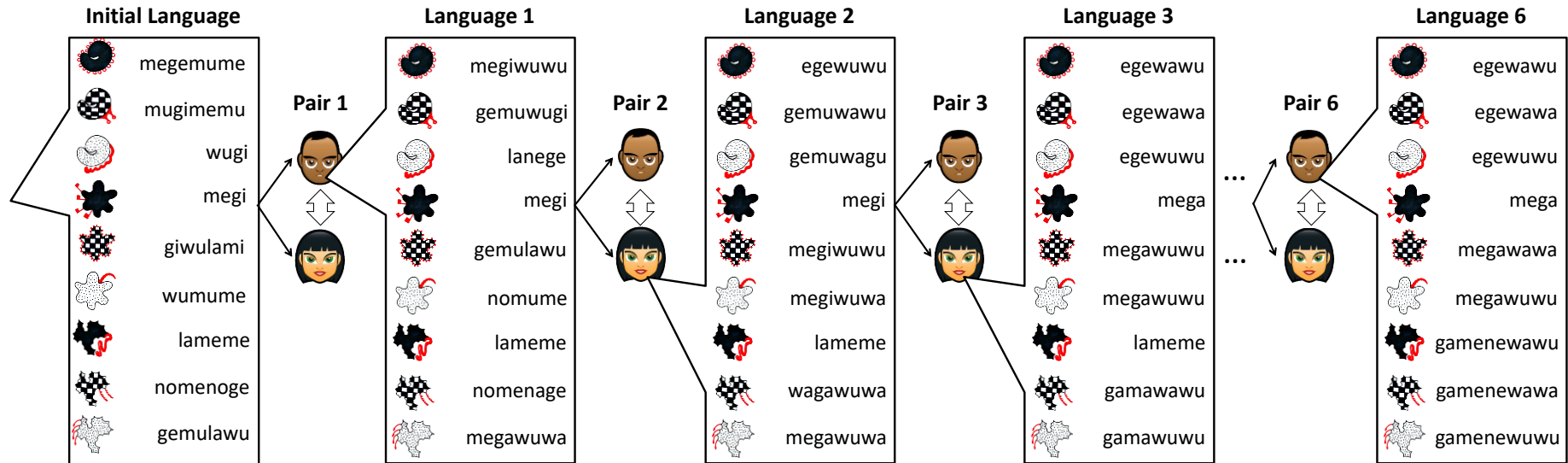
- The languages don't need to be **expressive**
- They get very simple

Can we add in a pressure for expressivity?

Kirby, Tamariz, Cornish & Smith (2015): Adding communication



Kirby, Tamariz, Cornish & Smith (2015): Adding communication, removing learning



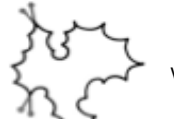
An initial language

 megemume	 megi	 lameme
 mugimemu	 giwulami	 nomenoge
 wugi	 wumume	 gemulawu
 lamege	 wulamugi	 megiwuwa

A final language from a chain

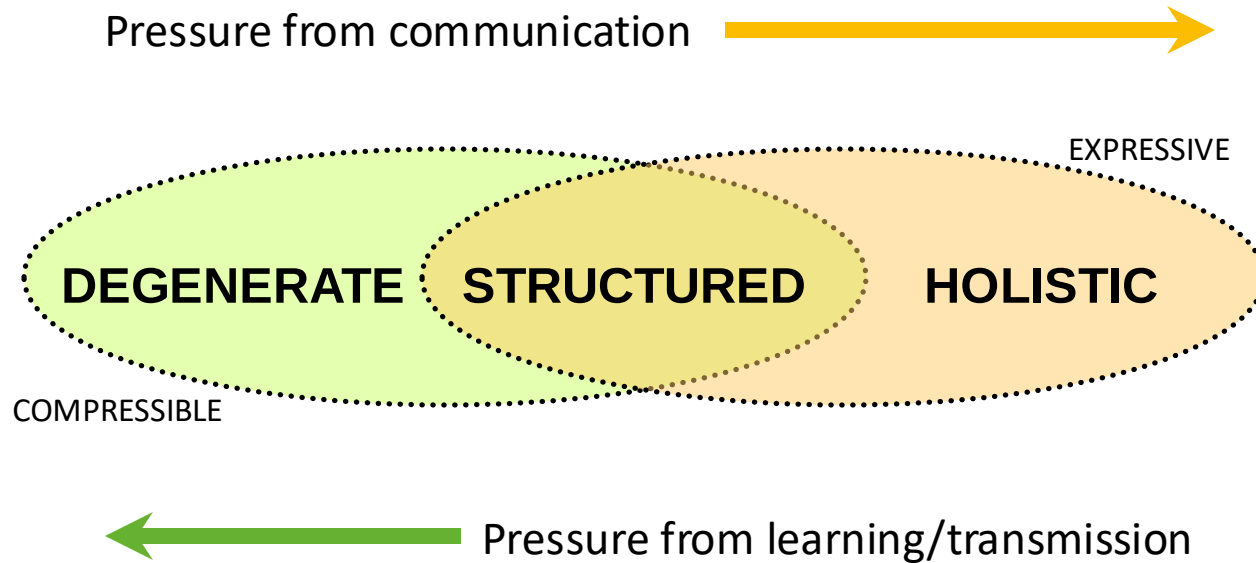
 egewawu	 mega	 gamenewawu
 egewawa	 megawawa	 gamenewawa
 egewuwu	 megawuwu	 gamenewuwu
 ege	 wulagi	 gamane

A final **holistic** language from a dyad

 manunumoko	 moko	 konu
 wekihumanunu	 mokowekihu	 lawa
 makihu	 mahiku	 wekihulawa
 manunumonu	 nomu	 wekihu

Learnability + expressivity = **structure**

Structure as a trade-off between compressibility and expressivity that plays out over cultural transmission



What about the beautiful adaptive fit between the structure of our thoughts and the structure of language?

Reminder from week 2

Pinker & Bloom (1990)



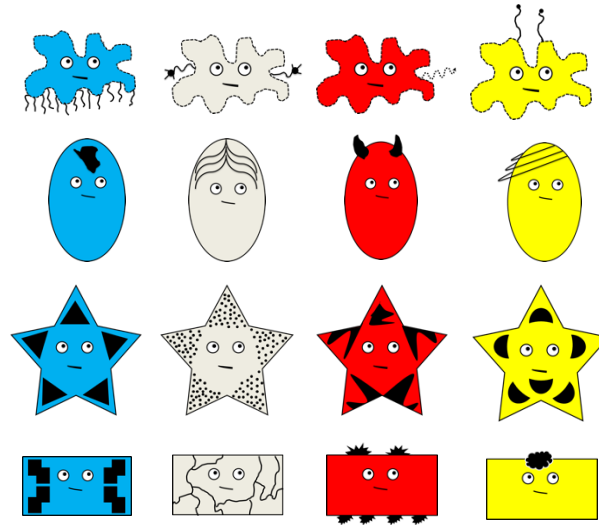
“All we have argued is that human language, like other specialized biological systems, evolved by natural selection. Our conclusion is based on two facts that we would think would be entirely uncontroversial: Language shows signs of complex design for the communication of propositional structures, and the only explanation for the origin of organs with complex design is the process of natural selection.” (p. 726)

Pinker, S., & Bloom, P. (1990). Natural language and natural selection. *Behavioural and Brain Sciences*, 13, 707-784.

 egewawu	 mega	 gamenewawu
 egewawa	 megawawa	 gamenewawa
 egewuwu	 megawuwu	 gamenewuwu
 ege	 wulagi	 gamane

The structure of the communicative task affects the kinds of structures that emerge

E.g. Winters, J., Kirby, S., & Smith, K. (2018). Contextual predictability shapes signal autonomy. *Cognition*, 176, 15-30.



The structure of the communicative task affects the kinds of structures that emerge

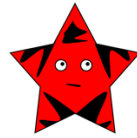
E.g. Winters, J., Kirby, S., & Smith, K. (2018). Contextual predictability shapes signal autonomy. *Cognition*, 176, 15-30.



kewa



lono



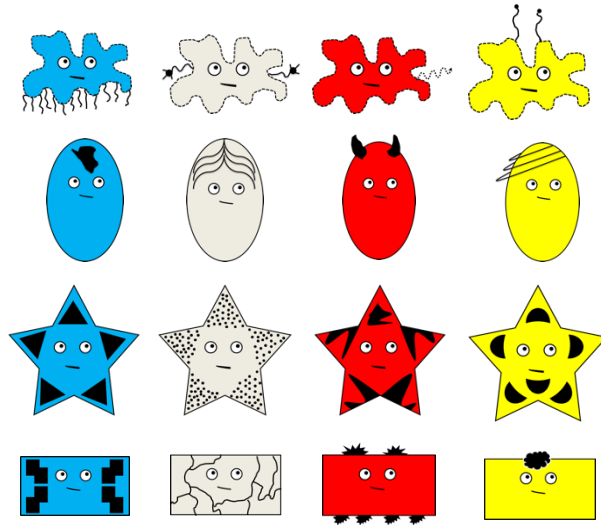
nunuki

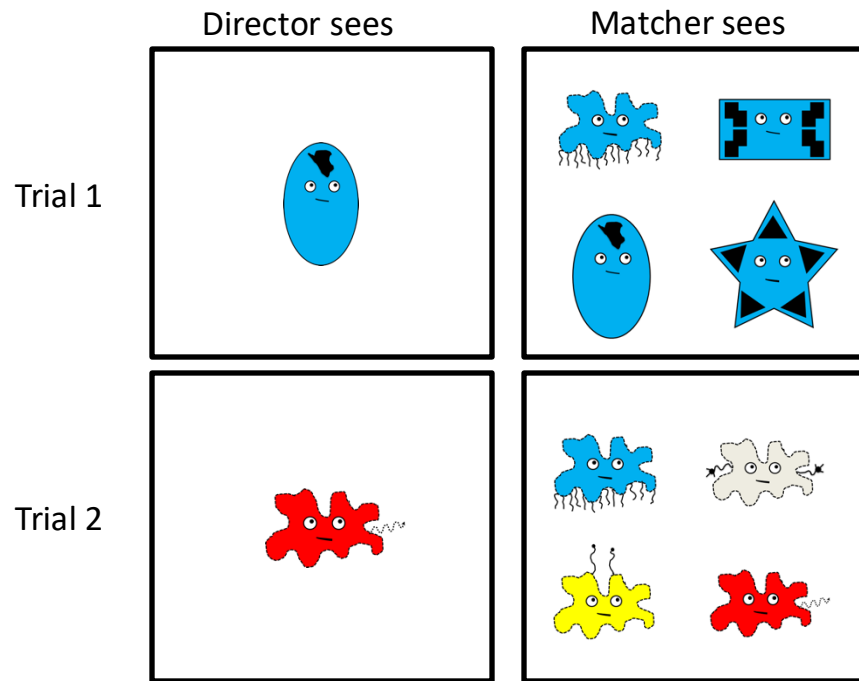


mopola

The structure of the communicative task affects the kinds of structures that emerge

E.g. Winters, J., Kirby, S., & Smith, K. (2018). Contextual predictability shapes signal autonomy. *Cognition*, 176, 15-30.

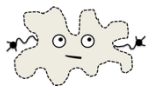




...



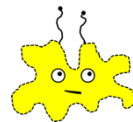
mege-ha



mege-hi



mege-hu



megi-lo



waka-ha



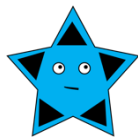
waka-hi



waka-hu



waki-lo



goko-ha



goko-hi



goko-hu



goki-lo



kuki-ha



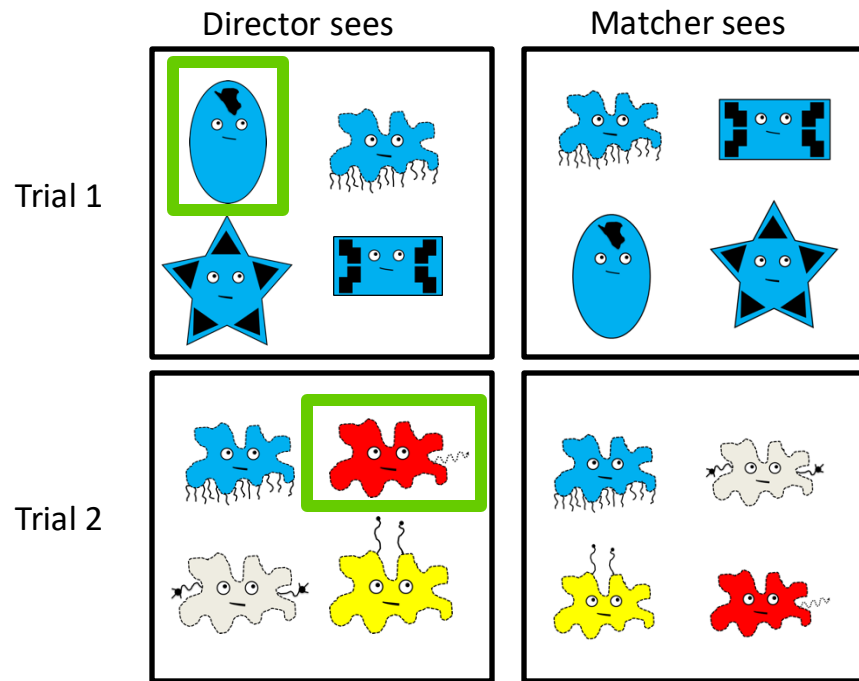
kuko-hi



kuko-hu



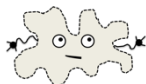
kuki-lo



On shape-relevant trials ...



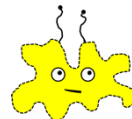
kewu



kewu



kewu



kewu



nunuki



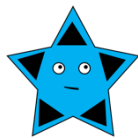
nunuki



nunuki



nunuki



lono



lono



lono



lono



mopola



mopola



mopola

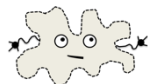


mopola

On colour-relevant trials...



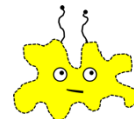
kewu mopola



kewu nunuki



kewu lono



kewu kewu



nunuki mopola



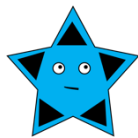
nunuki



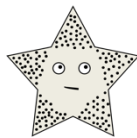
nunuki lono



nunuki kewu



lono mopola



lono nunuki



lono



lono kewu



mopola mopola



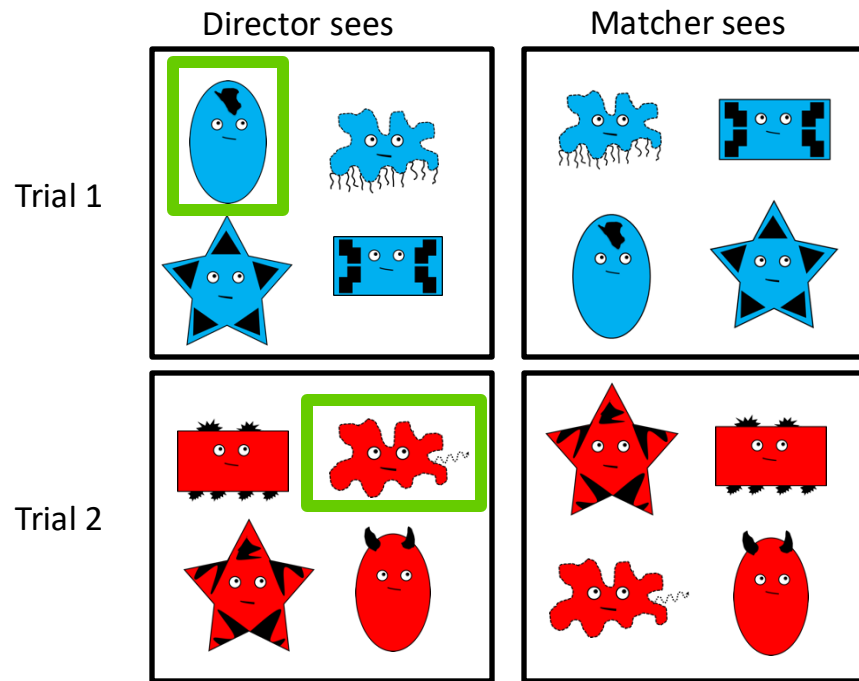
mopola nunuki



mopola lono

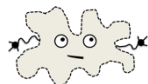


mopola kewu





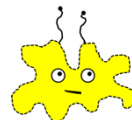
hagolo



hagolo



hagolo



hagolo



nuhumi



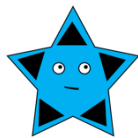
nuhumi



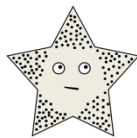
nuhumi



nuhumi



winigo



winigo



winigo



winigo



kamu



kamu



kamu



kamu

Reminder from week 2

Pinker & Bloom (1990)



“All we have argued is that human language, like other specialized biological systems, evolved by natural selection. Our conclusion is based on two facts that we would think would be entirely uncontroversial: Language shows signs of complex design for the communication of propositional structures, and the only explanation for the origin of organs with complex design is the process of natural selection.” (p. 726)

Pinker, S., & Bloom, P. (1990). Natural language and natural selection.
Behavioural and Brain Sciences, 13, 707-784.

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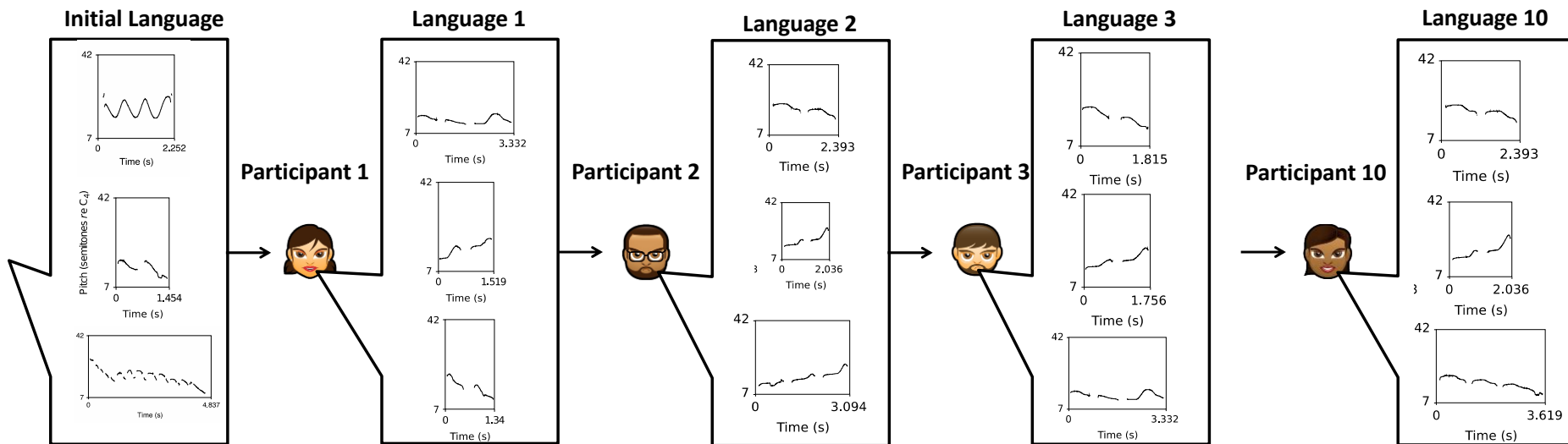
Example: duality of patterning

Language's communicative power comes from its **structure**

Duality of patterning: meaning-bearing units composed of (re)combinations of meaningless differentiating units

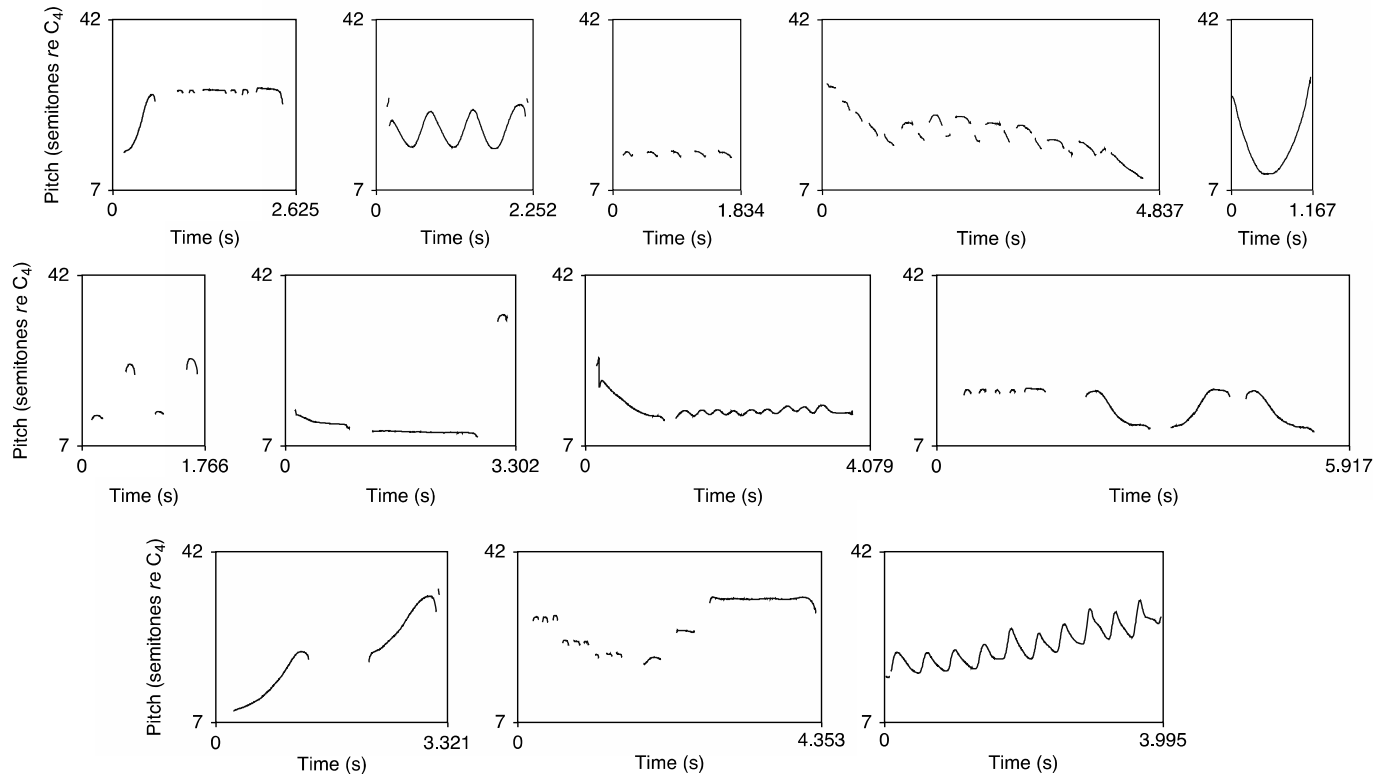
Word	Meaning
<i>log</i>	<i>"Noun; an unhewn portion of a felled tree"</i>
<i>dog</i>	<i>"Noun; A domesticated carnivorous mammal"</i>
<i>dig</i>	<i>"Verb; To work in making holes or turning the ground"</i>
<i>dim</i>	<i>"Adjective; Faintly luminous"</i>

Iterated Learning of Whistles

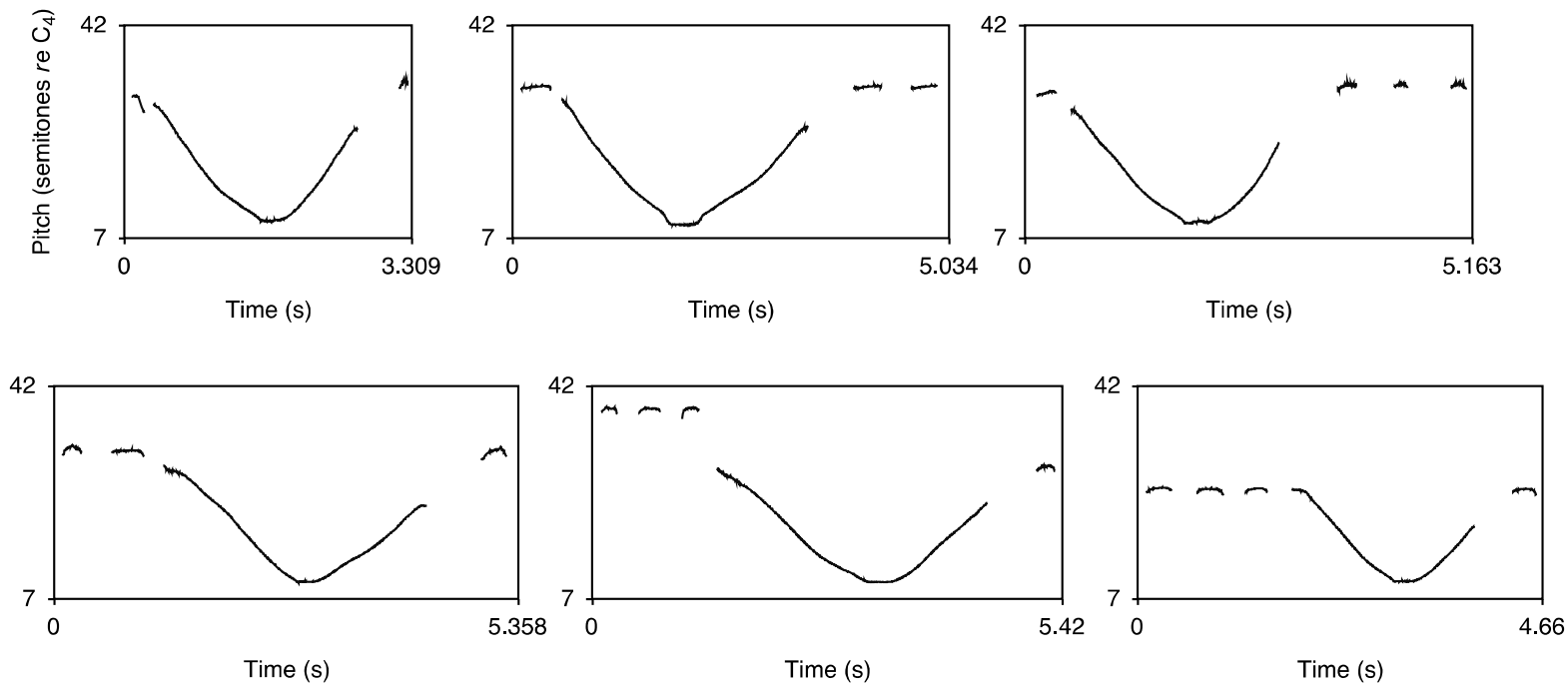


Verhoef, T., Kirby, S., & de Boer, B. (2014). Emergence of combinatorial structure and economy through iterated learning with continuous acoustic signals. *Journal of Phonetics*, 43, 57-68.

The initial whistle set



(Part of) A generation 10 whistle set



Cultural evolution of language: a summary

A uniformitarian approach

- We should attempt to explain the (hidden) past in terms of processes we can see operating in the present
- How far can we get in appealing only to the same processes we see shaping language in the present?

Language change

- (analogy-based) learning and (ostensive-inferential) use are important mechanisms

Language evolution

- Same processes can explain origins of symbols, compositionality, and duality of patterning
- At least in populations capable of the right kind of learning and use

Next up

- Tutorial
 - Do natural languages in different communities (transmitted under different constraints, with different communicative needs) show different adaptations to those different niches?
- Next week: guest lecture by Dr Annie Holtz
 - Sign language as a window into language origins
 - Reading plus pre-lecture quiz as per usual